

★ When height h given,

$$\begin{aligned} \text{min}^m \text{ no. of nodes needed to } (n) &= (h+1) \\ \text{generate Binary tree} \\ (n) \text{ Max}^m \text{ no. of nodes} &= (2^{h+1} - 1) \end{aligned}$$

★ If nodes are given,

$$\begin{aligned} \text{Min}^m \text{ height} &= \log_2 (n+1) - 1 \\ \text{Max}^m \text{ height} &= n - 1 \end{aligned}$$

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Height of a Binary Tree

$$\log_2 (n+1) - 1 \leq h \leq n - 1$$

No. of Nodes in a Binary Tree

$$h+1 \leq n \leq 2^{h+1} - 1$$